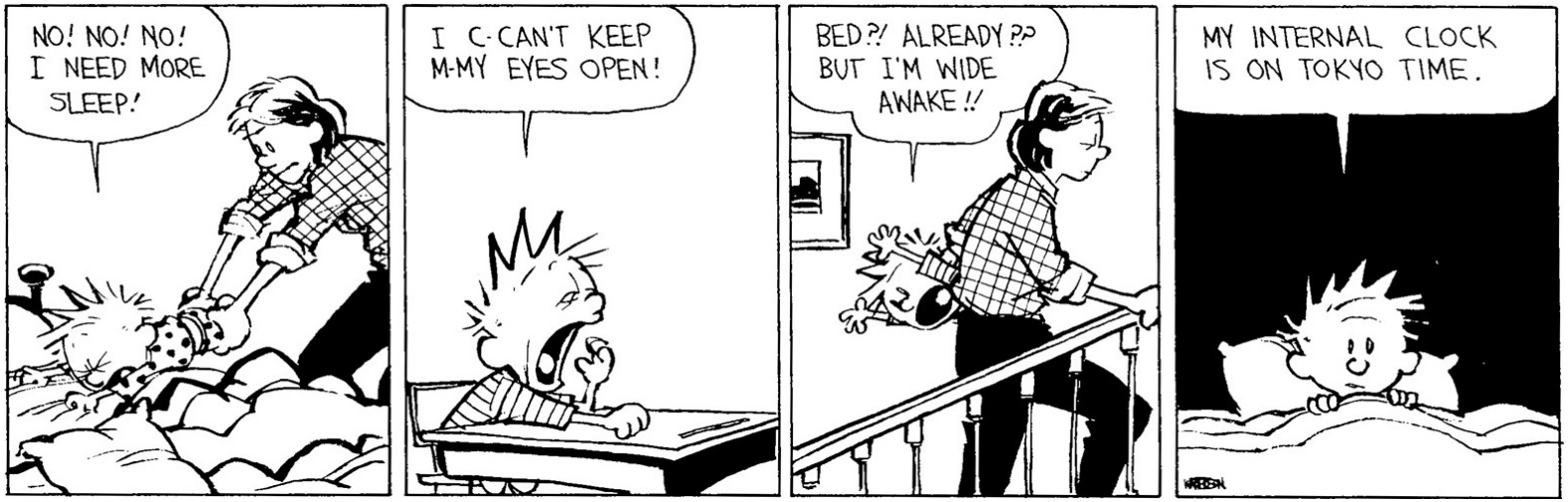


FATIGUE & SLEEP

Dr. Andrew Liu

ENP-0112

WHAT IS FATIGUE?

1999 US DOT Oper

Fatigue is *MORE THAN* sleepiness and its effects are more than falling asleep.

Fatigue is a complex state characterized by a lack of alertness and reduced mental and physical performance, often accompanied by drowsiness.

Fatigue may be caused or exacerbated by any or all of the following: lack of sleep, disruptive work/ rest cycles, neurological conditions, excess mental or physical workload, exposure to extreme physical conditions, emotional stress, the use of drugs or alcohol, illness, and/or monotony.

SLEEP DEPRIVATION AND FATIGUE PREVALENCE

- Humans used to go to bed at dusk and wake up at dawn
  - Electric light has allowed us to stay up later
  - Have jobs and schools shifted to accommodate us staying up later?
  - Technology has decreased the amount of time we sleep
- The percentage of adults averaging less than 6 hours per night of sleep has gone up by a factor of 10 in the last 50 years.
- The average hours of sleep of adults has reduced to 6.9 hours per night from 8.5 hours per night in the last 50 years.
- Hundreds of billions of dollars are spent or lost annually due to poor or limited sleep
  - Work and industrial accidents
  - Medication and counseling

Czeisler 2013; Colten and Altevogt, 2006

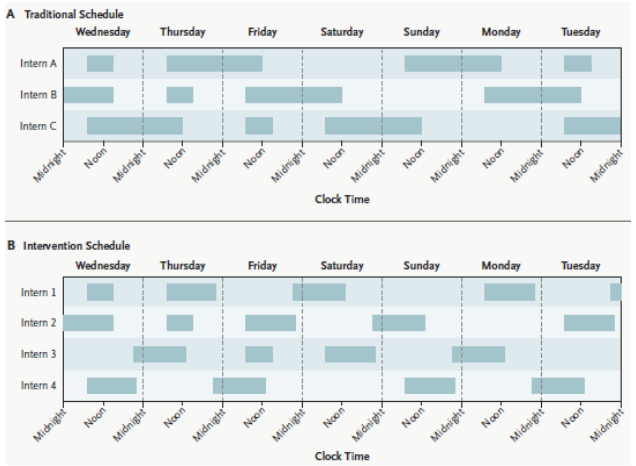
FATIGUE AND DRIVING

- Drowsy driving was responsible for 91,000 crashes, 50,000 injuries, and ~1200 deaths (NHTSA estimate, 2023)
- 2.1% of all fatal crashes were caused by a drowsy driver
- Occur most commonly with single driver and between midnight-6p and late afternoon

National Highway Traffic Safety Administration
AAA Foundation for Traffic Safety

FATIGUE AND MEDICAL ERRORS

- Traditional schedule
  - 77 - 81 hrs/week
  - Up to 34 continuous hours
- Intervention schedule
  - 60 - 63 hrs/week
  - Up to 16 consecutive hours



- Intervention schedule had:
  - Fewer serious medical errors (error in the delivery of medical care, whether harmful or trivial)
  - Fewer serious medication errors (error ordering or administering pharmaceutical agents, blood products, or intravenous fluids)
  - Fewer serious diagnostic errors (error related to history taking, performance of a physical examination, or ordering or interpretation of a diagnostic test)

Landrigan et al., 2004

WHAT FACTORS INFLUENCE SLEEP/FATIGUE?

- **Circadian rhythm**: a daily rhythm/oscillation that encourages and discourages sleep
- **Extended wakefulness**: continuous period of being awake that is longer than your natural circadian rhythm
- **Sleep loss (partial sleep restriction)**: regular sleep occurs, but each sleep session does not have enough hours
- **Sleep disorders**: disrupt sleep quality, e.g., sleep apnea

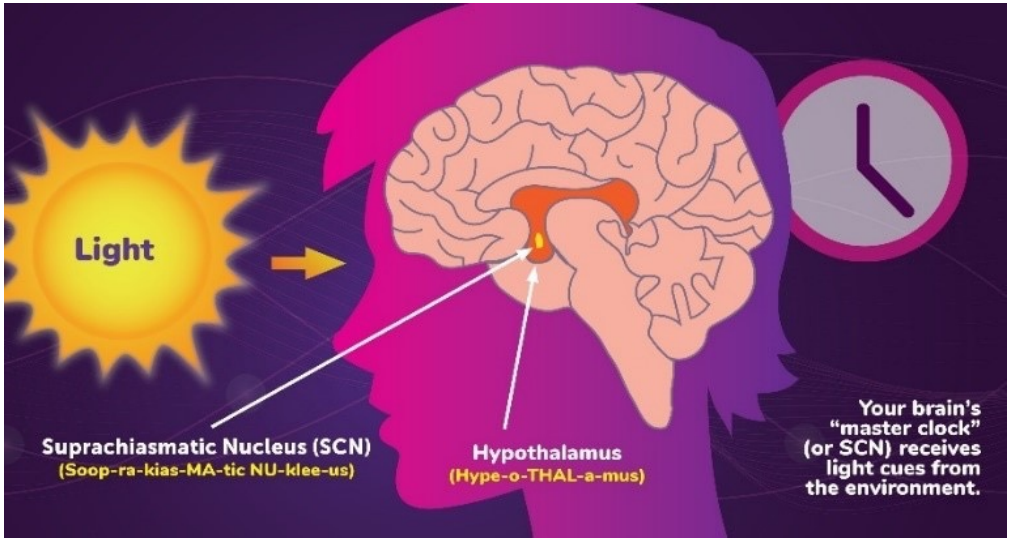
BORBÉLY TWO PROCESS MODEL OF SLEEP

- Process C (Circadian process):
  - Builds across the day and promotes wakefulness
  - Declines at bedtime to promote sleep
  - Builds during the night to promote waking up

- Process S (Sleep propensity/Homeostatic drive):
  - The need for sleep increases the longer you are awake
  - Dissipates with sleep
  - Correlates with the accumulation of adenosine

CIRCADIAN RHYTHM (PROCESS C)

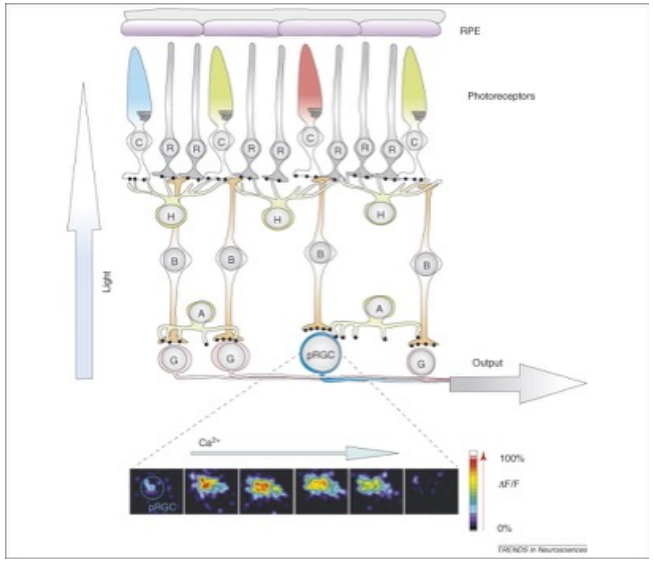
- A group of nerve cells in the hypothalamus called the suprachiasmatic nucleus (SCN) control the “master clock”
- Signal the production of melatonin
- Rhythms continue in the absence of light/dark cycles, BUT phase can be reset by light



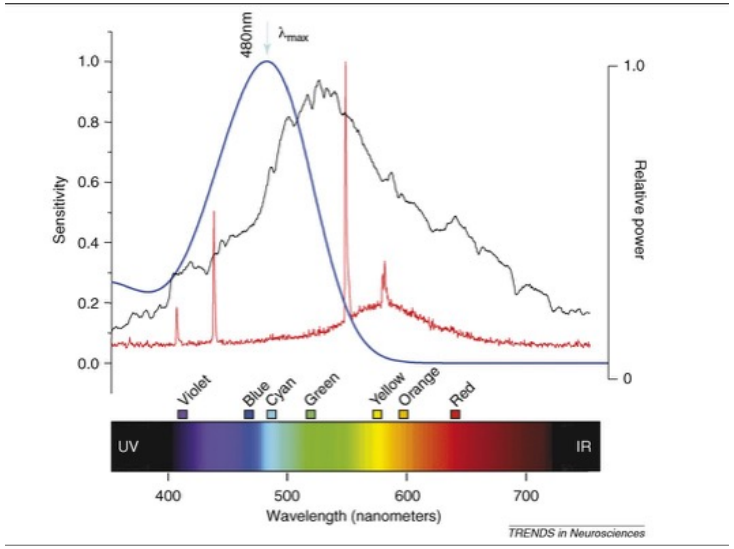
http://www.nigms.nih.gov/Education/Pages/Factsheet_CircadianRhythms.aspx

NONVISUAL CIRCADIAN PHOTOSENSITIVE CELLS

Photosensitive retinal ganglion cells regulate circadian rhythms

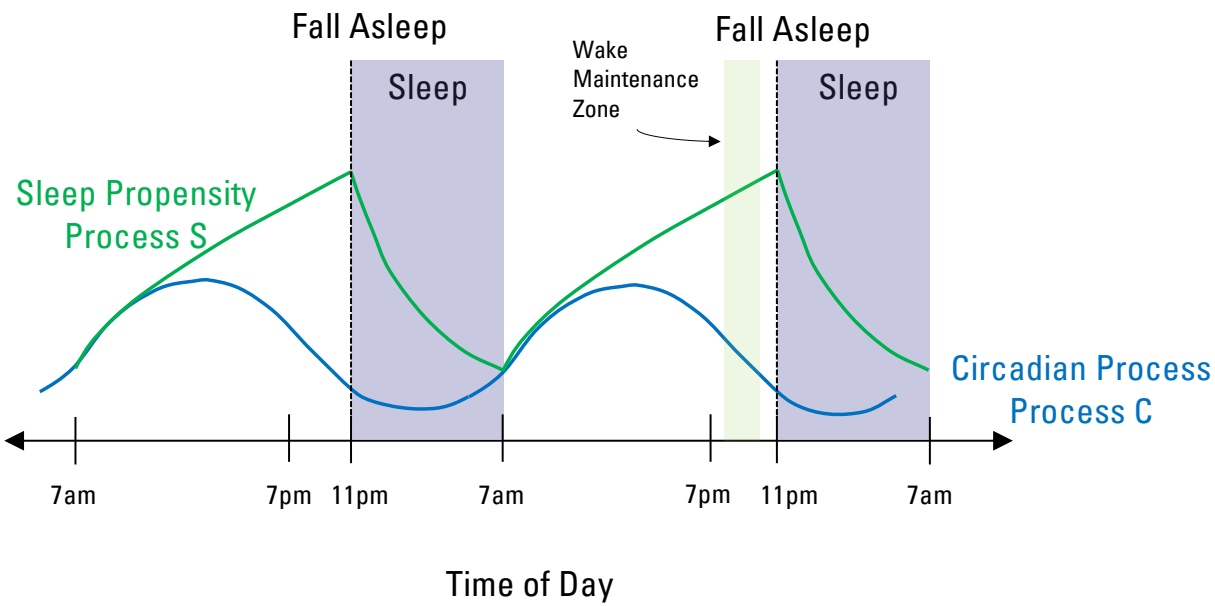


Photopigment melanopsin sends signals to the SCN

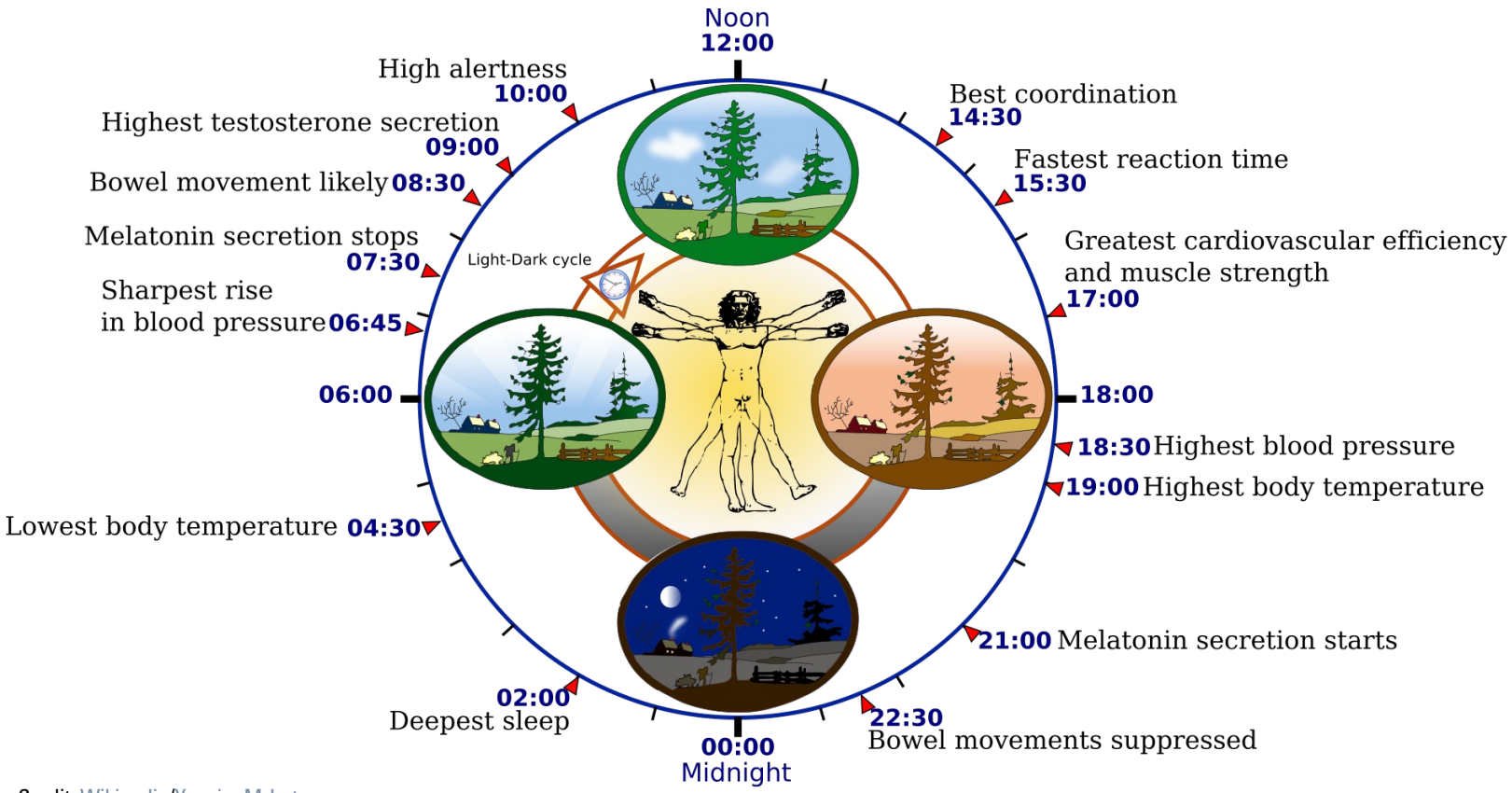


From Hankins, Peirson and Foster, 2008

SLEEP PROPENSITY (PROCESS S)

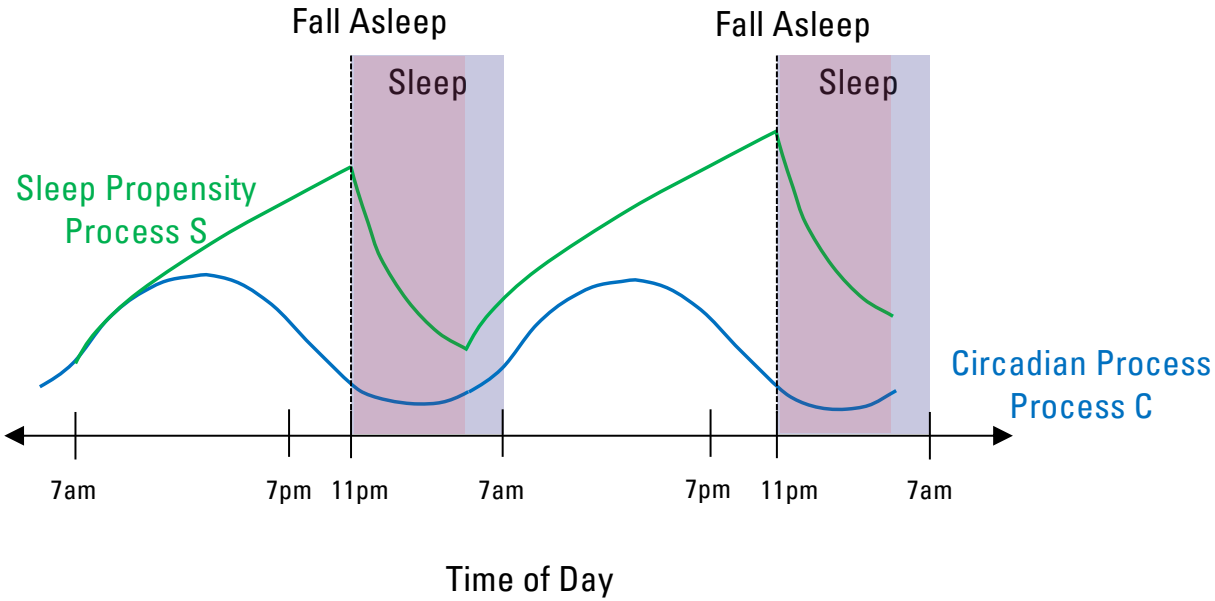


Adapted from Borbély, 1982



Credit: Wikipedia/YassineMrbat

EFFECT OF SLEEP RESTRICTION



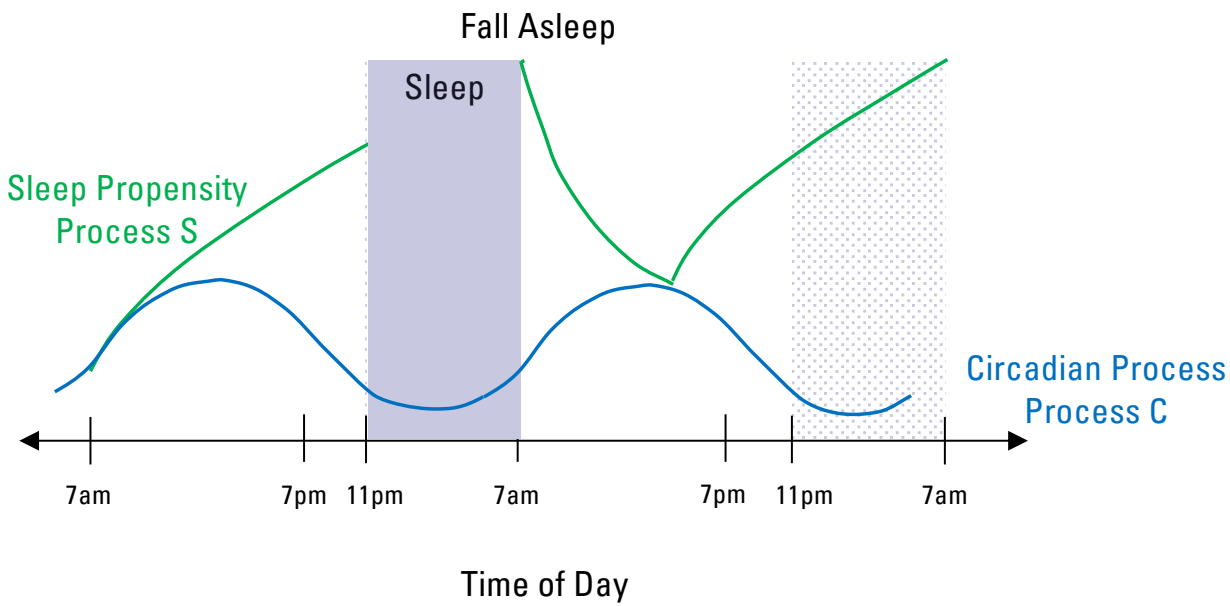
Adapted from Borbély, 1982

RISKS OF SLEEP DEBT

- Nearly 1/4 of all workers have shifts that are not during the daytime
    - >2/3's of these workers have problem sleepiness and/or difficulty sleeping.

- Sleep deprivation slows reactions to stimuli, decreases accuracy of responses, and leads to long lapses in attention
    - People who slept 2 hours or less within 24-hours are not fit to operate a vehicle
    - Drivers who usually sleep 4-5 hours/day
        - had 5.4 times the crash rate of drivers who sleep for >7 hours
          https://aaafoundation.org/acute-sleep-deprivation-risk-motor-vehicle-crash-involvement/
        - Crash risk is similar to risk of driving with BAC of 0.08 or higher
    - Crash risk < 4 hours sleep is comparable to risk of driving with BAC of 0.12-0.15

EFFECT OF SHIFT WORK



Adapted from Borbély, 1982

EFFECTS OF SHIFT WORK

- ~15 M Americans work full time (32-40 hours) on evening shift, night shift, rotating shifts, or other employer-arranged irregular schedules
  - Out of 132M full-time workers (Bureau of Labor Statistics, 2019)
  - https://www.cdc.gov/niosh/work-hour-training-for-nurses/default.html
  - https://www.sleepfoundation.org/articles/how-atypical-work-schedules-affect-performance

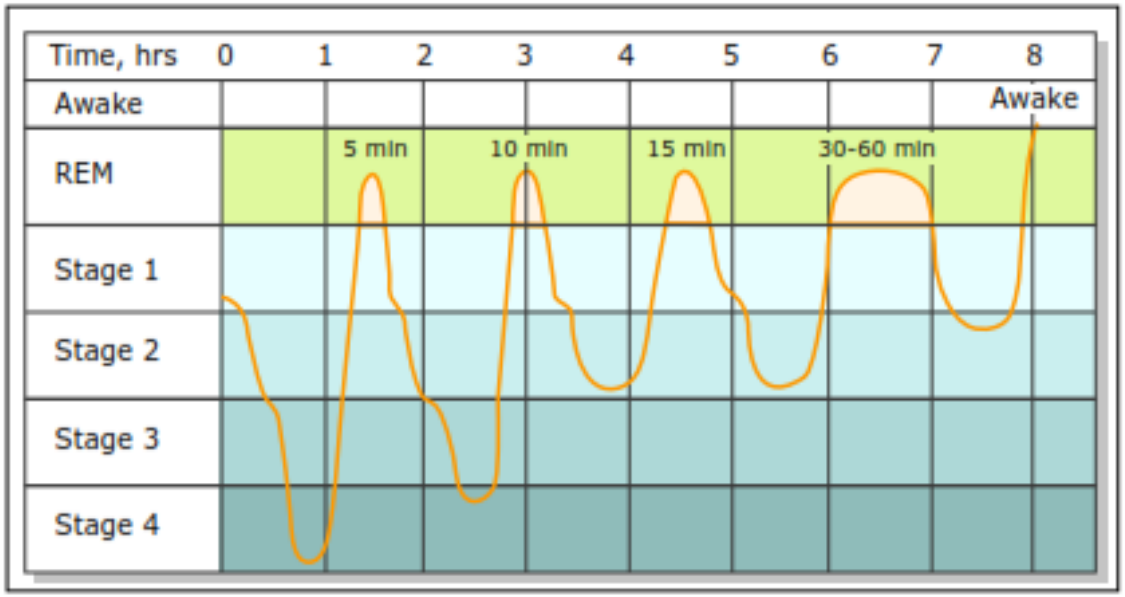
- 19% of working adults work 48 hours or more per week, 7% worked 60 hours (2010)
- 2 out of 5 work non-standard shifts
  - https://www.prb.org/workingaroundtheclock/

- Shift work and long work hours are associated with health and safety risks
  - https://onlinelibrary.wiley.com/toc/10970274/2022/65/11

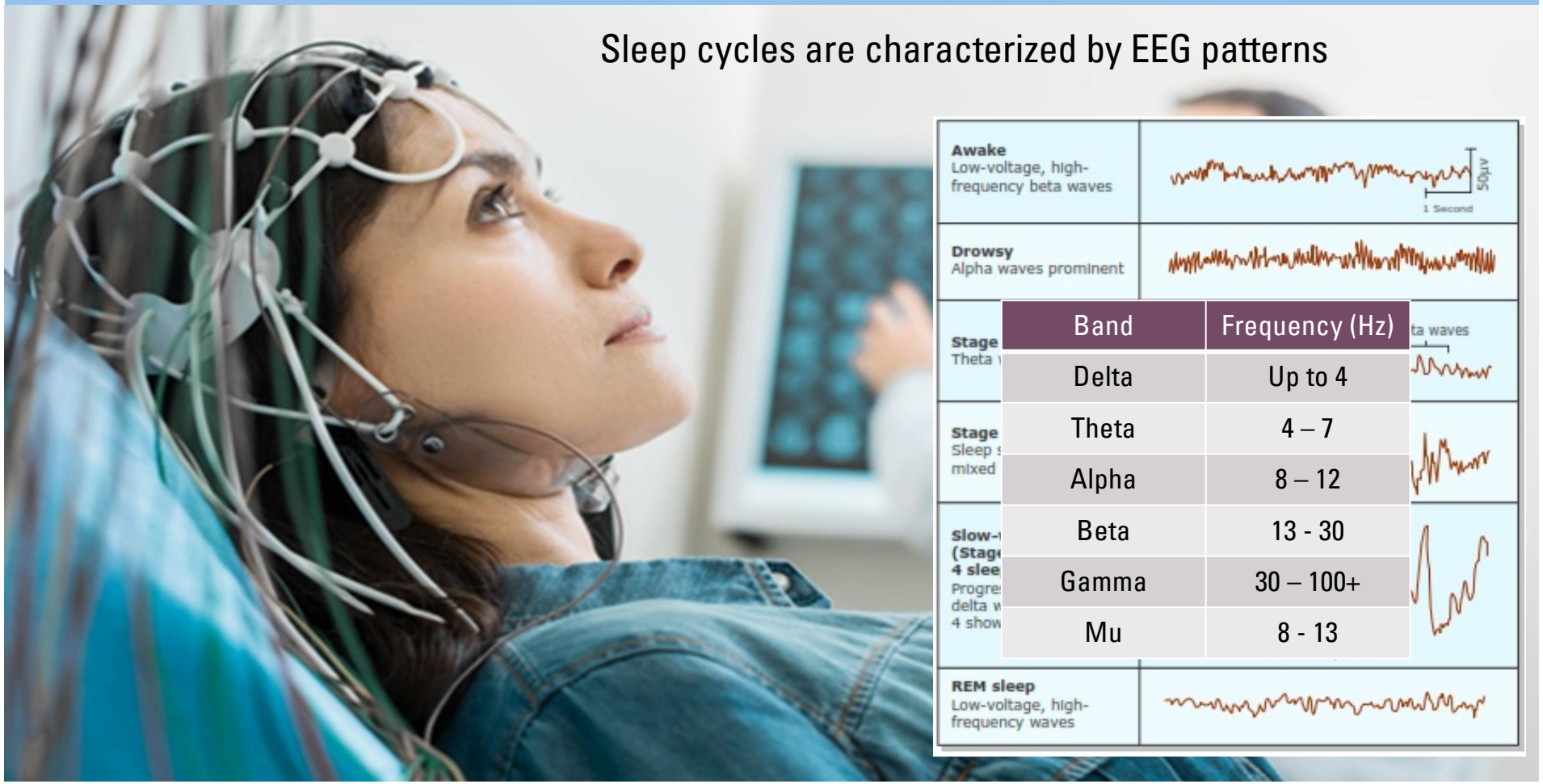
TYPES OF SLEEP

- Sleep quality is cyclical, with many stages and depths
  - Non-rapid eye movement (NREM)
  - Rapid eye-movement (REM)

TYPICAL SLEEP PATTERN OF A HEALTHY ADULT



Sleep cycles are characterized by EEG patterns



Band	Frequency (Hz)
Delta	Up to 4
Theta	4 – 7
Alpha	8 – 12
Beta	13 - 30
Gamma	30 – 100+
Mu	8 - 13

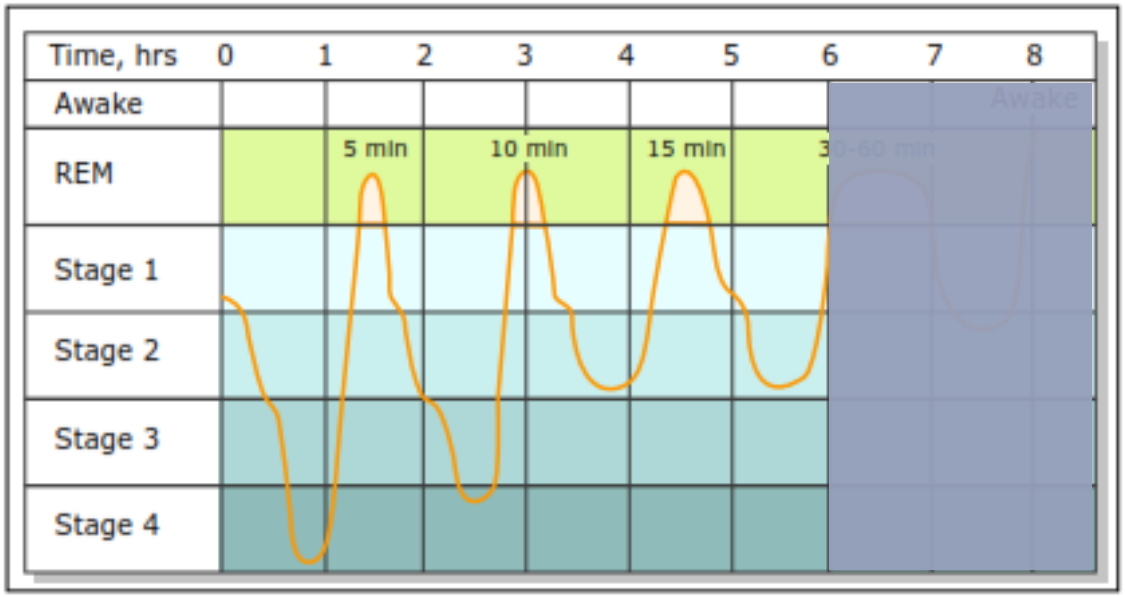
IMPORTANCE OF SLEEP CYCLES

- Without Slow wave sleep (Levels 3 and 4), feel unrested, reduction in cognitive performance

- Without REM sleep, have more moodiness, hypersensitivity, inability to consolidate complex learning.

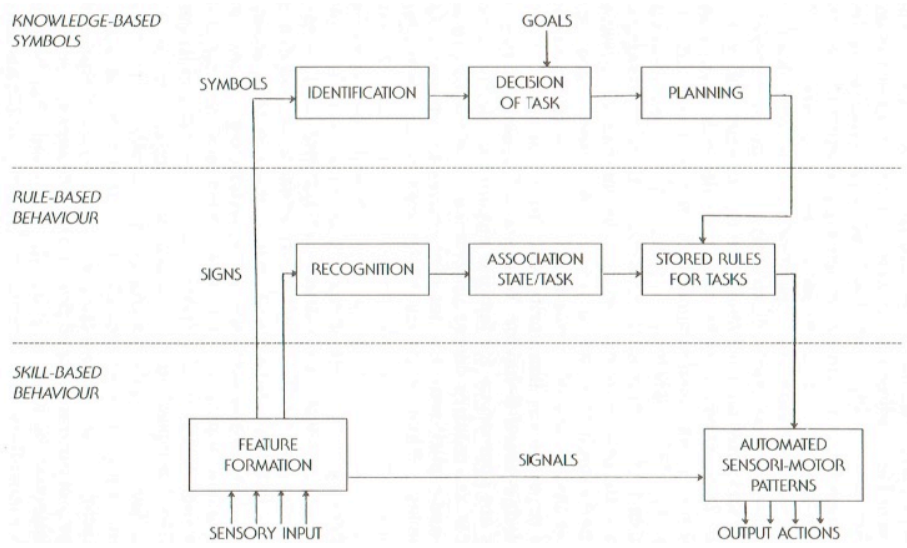
REM sleep is important for psychological well-being!

TYPICAL SLEEP PATTERN OF A HEALTHY ADULT



EFFECTS OF FATIGUE – LOWER LEVEL

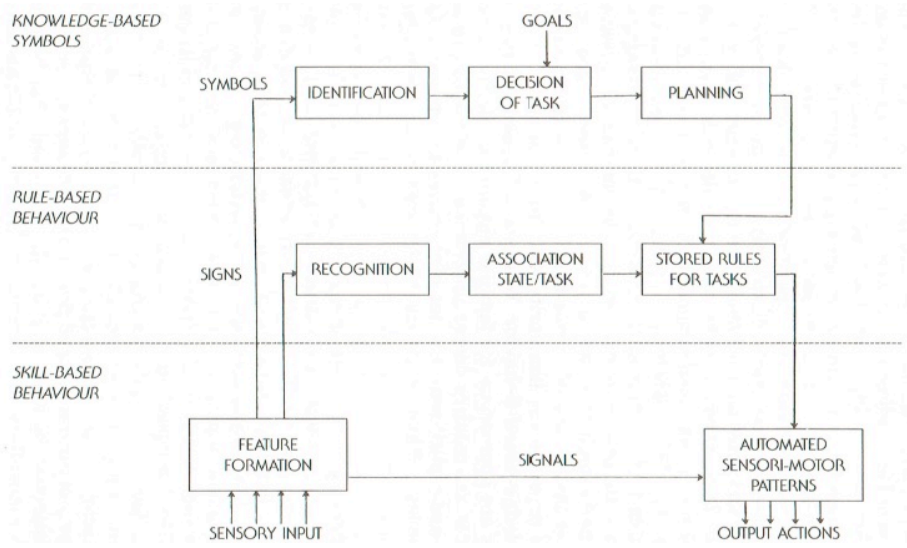
- Slower reaction times
- Reduced response accuracy
- Attentional lapses
- Increased task performance time
- Increased number of errors
- Increased unnecessary movements
- Omission of details
- Increased distractibility



Rasmussen SRK model, 1983

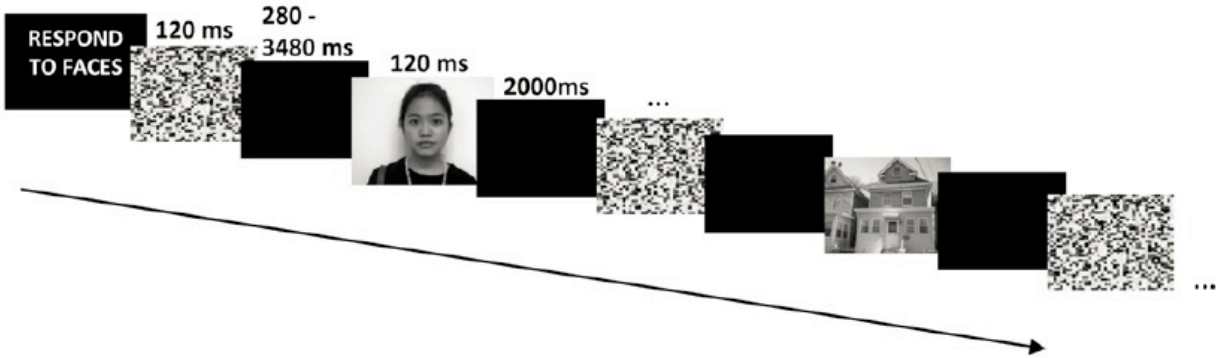
EFFECTS OF FATIGUE – HIGHER LEVEL

- Decreased executive attention and executive control
- Decreased working memory
- Sub-optimal planning
- Impaired decision making, ability to handle competing distractions
- Impaired innovation, ability to revise plans
- Less effective communication



Rasmussen SRK model, 1983

EFFECTS OF FATIGUE ON SELECTIVE ATTENTION

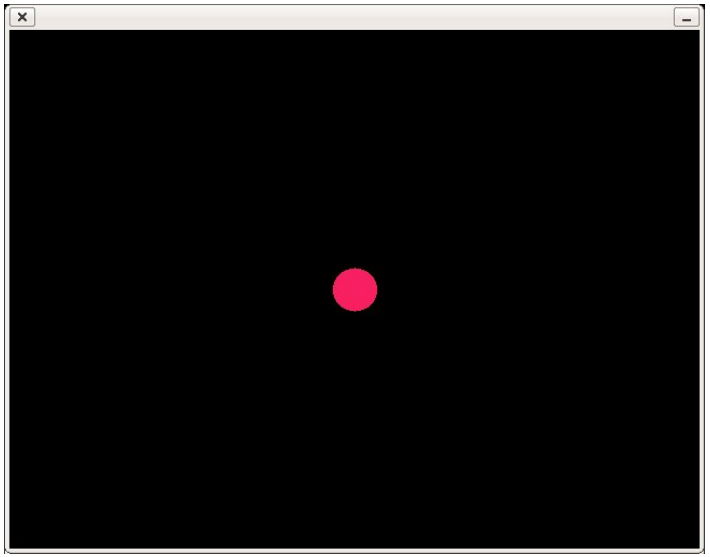


- Examined the ability to respond to pictures of faces, houses, and scrambled image when rested and sleep deprived
- Participants asked to “Attend to House” or “Attend to Face” and responded when the correct image appeared
- Reaction times were not significantly different between groups, but accuracy decreased on average from 91% to 75%

Lim et al., 2010

MEASURING SLEEPINESS OR FATIGUE

Objective: Psychomotor Vigilance Test (PVT)

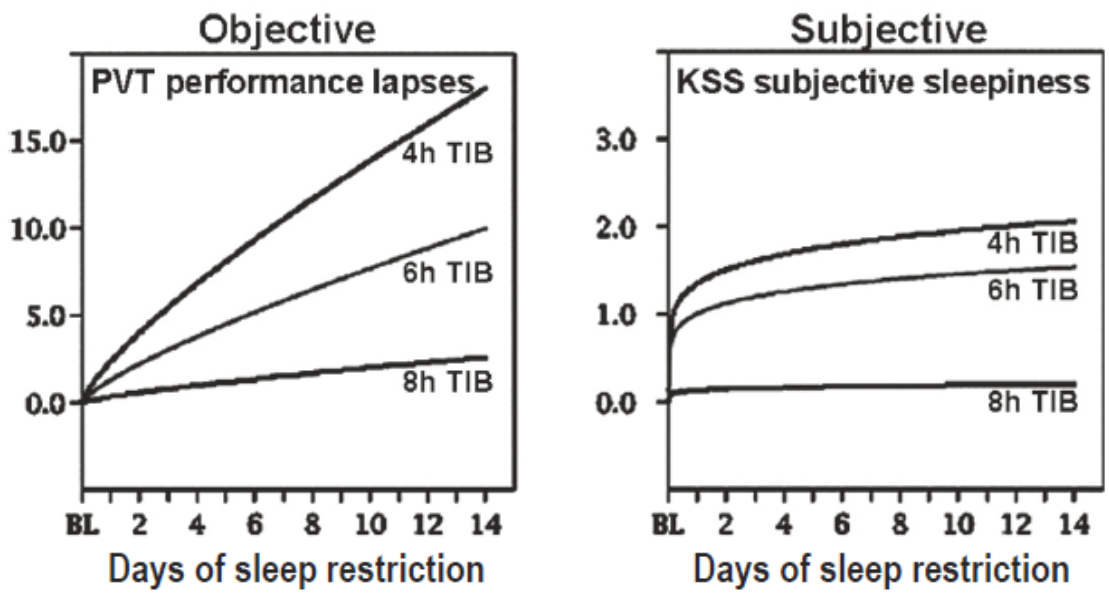


Subjective: Karolinska Sleepiness Scale (KSS)

Please indicate your sleepiness during the 5 minutes before this rating by checking the box next to the appropriate number.
Use also the intermediate steps!

- ☐ 1 –very alert
- ☐ 2 –
- ☐ 3 – alert – normal level
- ☐ 4 –
- ☐ 5 – neither alert nor sleepy
- ☐ 6 –
- ☐ 7 – sleepy – but no effort to keep awake
- ☐ 8 –
- ☐ 9 – very sleepy, great effort to keep awake, fighting sleep

PERCEPTION OF SLEEPINESS



Banks and Dinges, 2007

FATIGUE COUNTERMEASURES

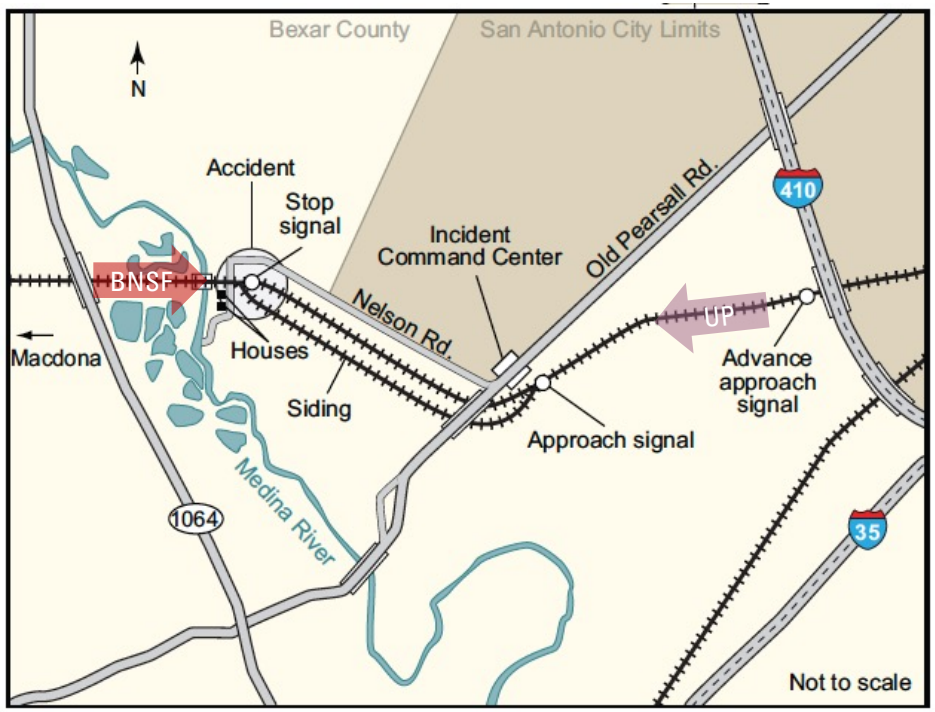
- Individual
  - Sleep
  - Caffeine
  - Lighting
  - Physical exercise
- Organization
  - Scheduling
  - Physical environment
  - Fatigue management training

UP-BNSF TRAIN COLLISION, MACDONA, TX (JUNE 28, 2004)

NTSB Accident report RAR-06/03
(https://www.ntsb.gov/investigations/AccidentReports/Reports/RAR0603.pdf)

ACCIDENT SYNOPSIS

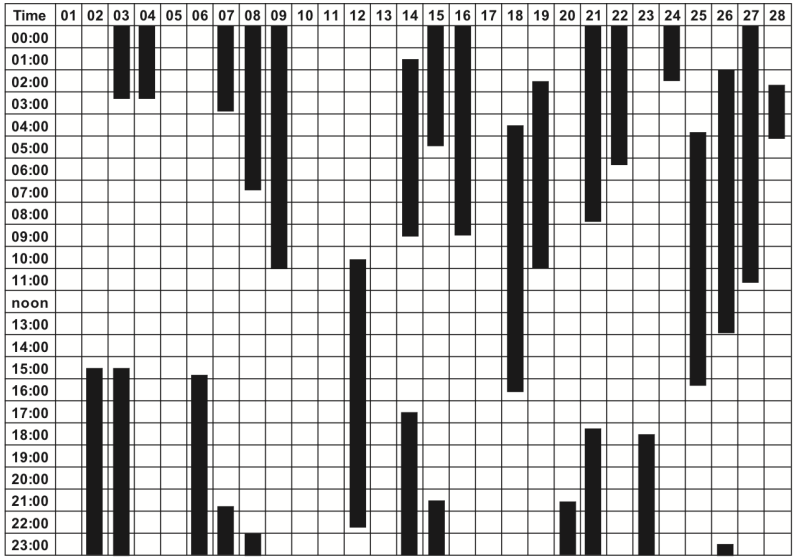
- Eastbound BNSF train instructed en route to pull onto siding for relief crew.
- Westbound UP train passed ADV APPR signal at 25mph but engineer increased speed after passing signal.
- UP train passed APPR signal at 46mph
  - Passed head of BNSF train on siding but didn't dim lights.
- Accident occurred at 5:00a
  - UP train passed the STOP signal and struck the 63rd car of the BNSF train



UP CREW WORK, REST, AND FATIGUE

- UP engineer schedule
  - Work periods in June were 9-18+ hrs time on duty.
  - Obstacles to restorative sleep
    - Intermittent day and night schedules created
    - Did not sleep at his own residence
- UP conductor schedule
  - Established a nighttime sleep pattern in the preceding 10 days
    - Call for the accident trip was at midnight!
    - Failed to get adequate sleep the night before the accident trip (only a couple hours)

Table 1. UP engineer's work history June 1-28, 2004.



Black bars indicate hours on duty.

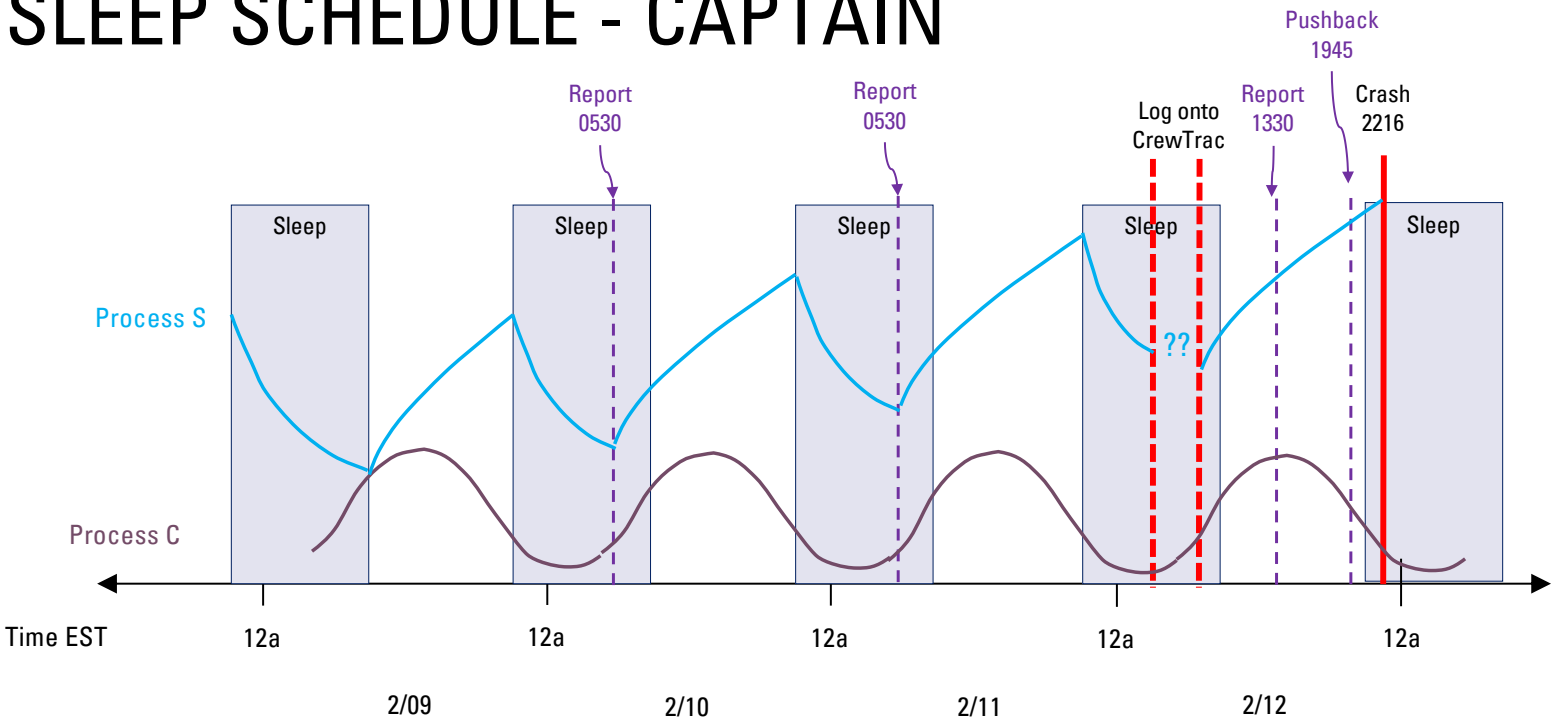
EVIDENCE OF ERRORS DUE TO FATIGUE

- [Engineer] Failed to blow the horn when passing through rail/highway grade crossings (2x) before ADV APPR signal
- [Engineer] Inappropriate control actions prevented the alerter system from sounding and stopping the train.
  - Sped up after ADV APPR; Did not slow after APPR;
  - Did not dim headlights when passing another train

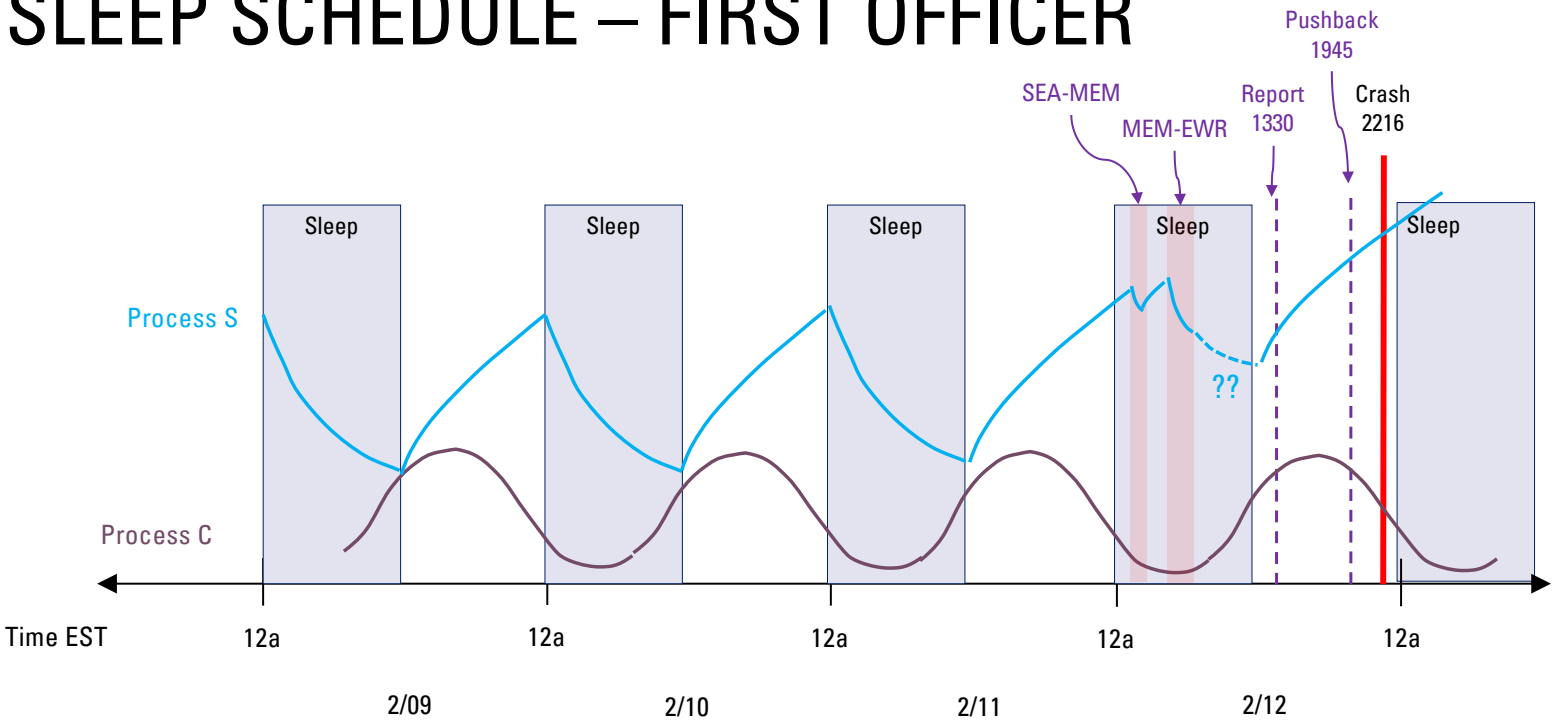
- [Conductor] Failed to acknowledge and record restrictive signals (non-green) prior to the accident
- [Conductor] Failed to correct errors by the engineer (e.g., blowing the horn)
- [Conductor] Post-accident, asked engineer “What happened?”

REVIEW OF COLGAN AIR

SLEEP SCHEDULE - CAPTAIN



SLEEP SCHEDULE – FIRST OFFICER



PILOT ACTIONS RELATED TO FATIGUE

- Low level
  - Attentional lapses – failure to detect impending stick shaker activation (speed)
  - Slower response time –
  - Increased distraction -

- High level
  - Improper response to the stick shaker.
  - Retraction of flaps and landing gear.
  - Task management and prioritization

TIPS FOR GOOD SLEEPING HABITS

- Sleep must be a consistent priority
- Avoid caffeine later in the day
- Sleep in a very dark, quiet and cool place free of light from television or cell phones
- Turn off all unnecessary light after dusk
- Consistently go to sleep at approximately the same time, including weekends
- Naps may help or hurt your sleeping situation
  - If you are very sleep deprived, naps may help in sleep amount without hurting your ability to fall asleep at night
  - If you have difficulty falling asleep at night, naps will reduce your sleep drive.

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